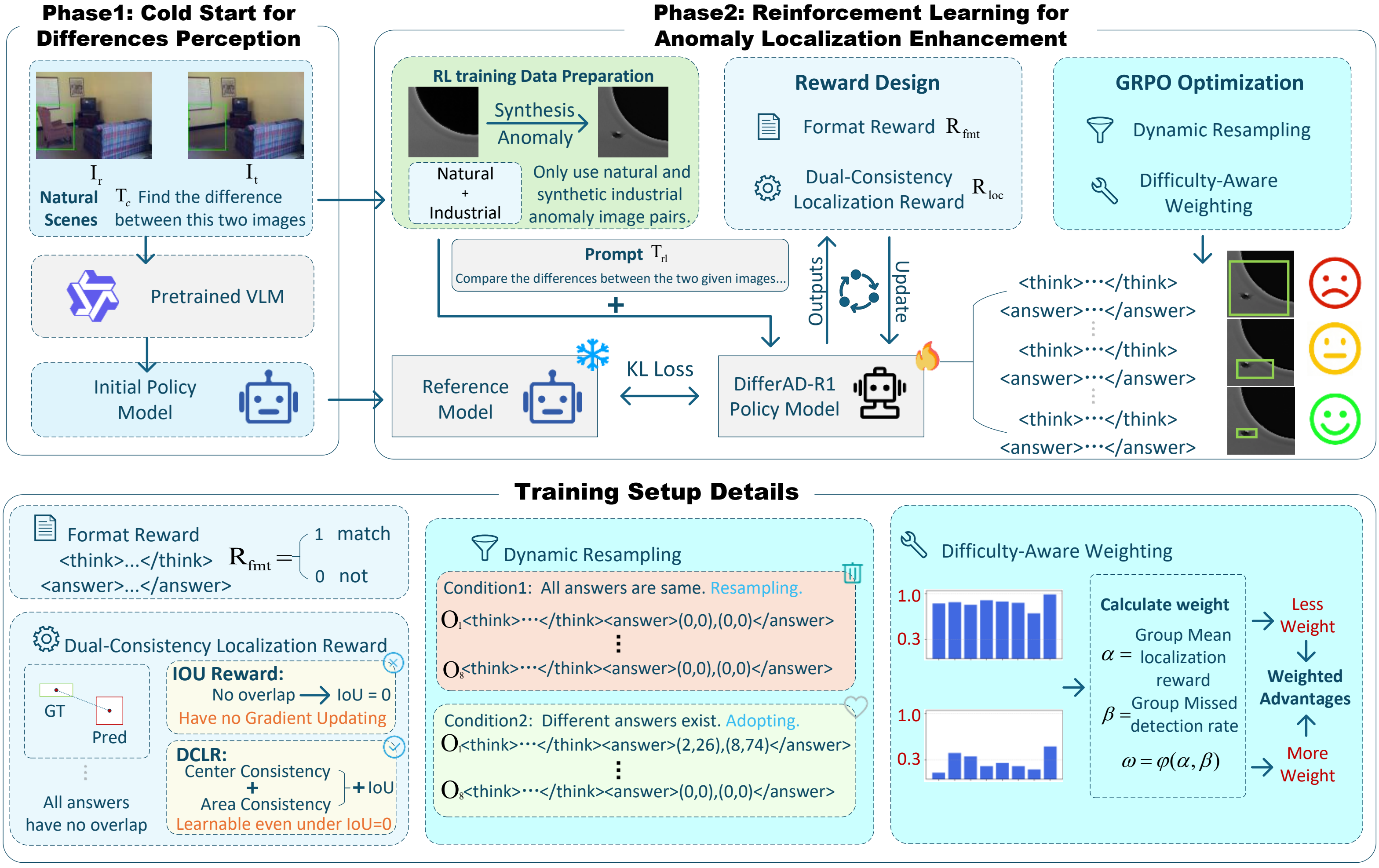

Phase1: Cold Start for Differences Perception
I_r
I_t
Natural Scenes
T_c Find the difference between this two images
Pretrained VLM
Initial Policy Model
Phase2: Reinforcement Learning for Anomaly Localization Enhancement
RL training Data Preparation
Synthesis Anomaly
Natural + Industrial
Only use natural and synthetic industrial anomaly image pairs.
Reward Design
Format Reward R_fmt
Dual-Consistency Localization Reward R_loc
GRPO Optimization
Dynamic Resampling
Difficulty-Aware Weighting
Prompt T_rl
Compare the differences between the two given images...
Outputs
Update
Reference Model
KL Loss
DifferAD-R1 Policy Model
<think>···</think>
<answer>···</answer>
Training Setup Details
Format Reward
<think>...</think>
<answer>...</answer>
R_fmt = 1 match, 0 not
Dual-Consistency Localization Reward
GT
Pred
All answers have no overlap
IOU Reward:
No overlap → IoU = 0
Have no Gradient Updating
DCLR:
Center Consistency + Area Consistency + IoU
Learnable even under IoU=0
Dynamic Resampling
Condition1: All answers are same. Resampling.
O_1<think>···</think><answer>(0,0),(0,0)</answer>
O_8<think>···</think><answer>(0,0),(0,0)</answer>
Condition2: Different answers exist. Adopting.
O_1<think>···</think><answer>(2,26),(8,74)</answer>
O_8<think>···</think><answer>(0,0),(0,0)</answer>
Difficulty-Aware Weighting
1.0
0.3
Calculate weight
α = Group Mean localization reward
β = Group Missed detection rate
ω = φ(α, β)
Less Weight
Weighted Advantages
More Weight